HealthConnect Rwanda

Healthcare operations and performance analytics

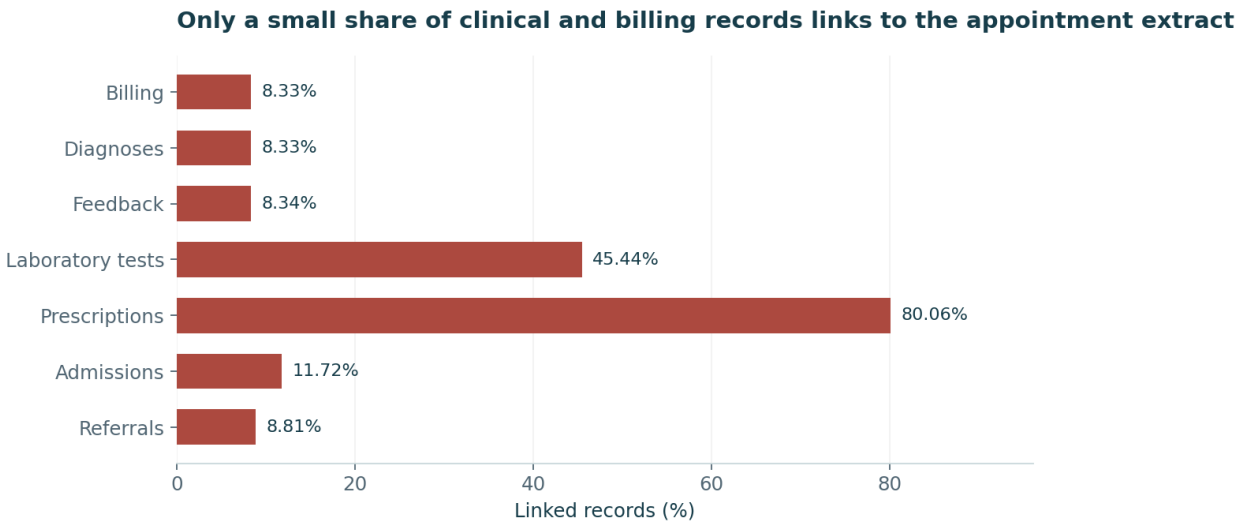
Business insights report

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The supplied 2026 scenario contains 20,000 registered patient records, 5,000 available appointments, 60,000 bills and 21,064 feedback responses. The principal constraint is incomplete encounter linkage: 55,000 bills have no matching appointment in the supplied extract. Full financial totals and linked facility comparisons therefore answer different questions.

Measure	Result	Reporting basis
Registered patients	20,000	Patient master, all records
Observed appointment patients	4,443	Distinct patients in 5,000 available appointments
Recorded payment amounts	RWF 8,389,454,556	All 60,000 payment records; dates absent
Positive bill balances	RWF 919,947,405	Billed less aggregated paid, bounded at zero
Mean feedback rating	2.99 / 5	All 21,064 supplied responses

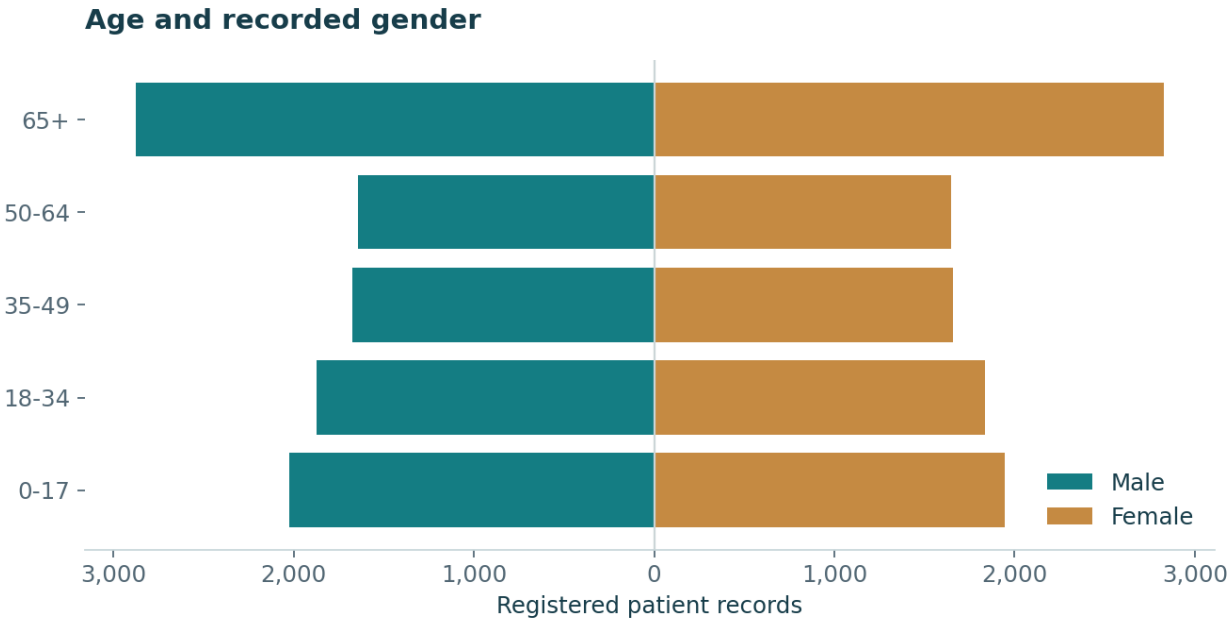
Coverage governs interpretation



Source: supplied HealthConnect files. The annual scenario extends to 28 December 2026; it includes 864 appointments dated after 25 October. These records are analysed as supplied scenario data, not verified completed clinical activity.

Patient register and clinical activity

The register is 50.45% Male and 49.55% Female, with a mean supplied age of 45.03 years. Every record is labelled Active and carries an insurance code. These fields do not verify recent attendance or policy validity. Only 4,443 registered patients appear in the available appointments; 510 have at least two observed appointments.



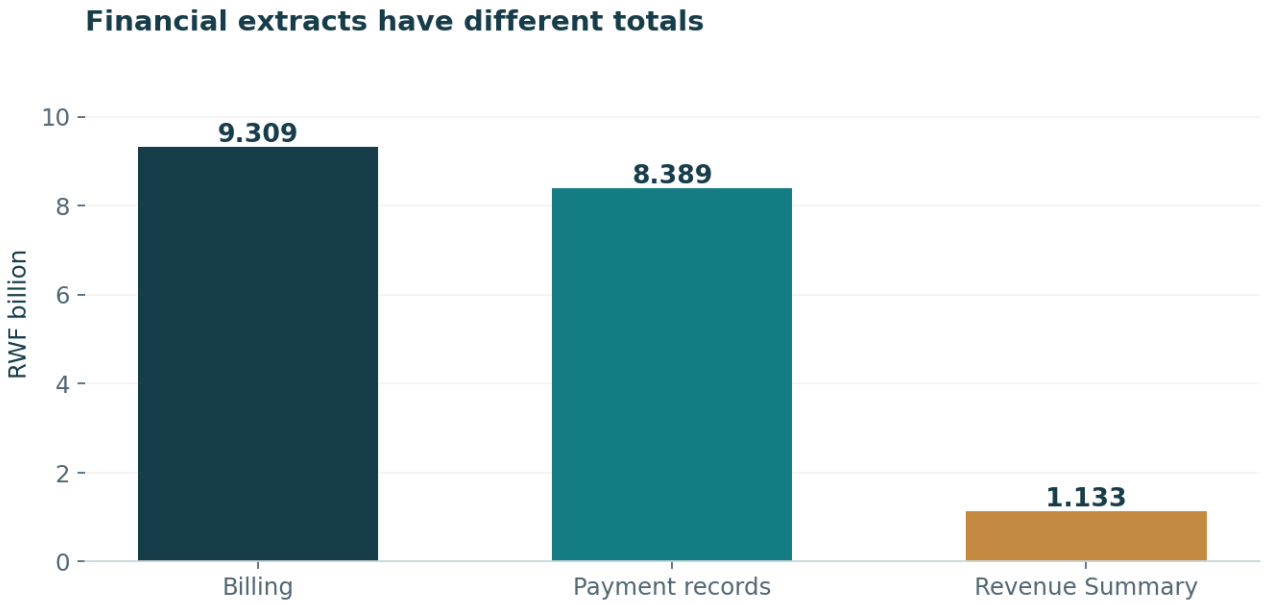
Recorded chronic conditions appear on 14,929 patient records (74.645%). Asthma, hypertension and diabetes are descriptive register categories; no clinical diagnosis or treatment recommendation is inferred from these fields.

Clinical extracts use different reporting grains

Extract	Available records	Interpretation
Diagnoses	60,000	Full CSV selected; workbook's 5,000 rows are an identical subset
Laboratory tests / prescriptions	5,000 / 5,000	Full event counts retained; only matched appointments support date/facility attribution
Admissions	5,000	Supplied mean length of stay: 5.49 days
Referrals	4,708	Destination identifiers available; origin requires appointment linkage

New registration counts, cancellation rates and clinical readmission rates are unavailable. Registration dates, appointment statuses, and dated admission/discharge episodes are absent. An observed repeat-admission count is reported separately and is not a 30-day readmission rate.

Financial position and reconciliation



Billing totals RWF 9,309,401,961; payment records total RWF 8,389,454,556. The difference is RWF 919,947,405, with no calculated credit balances. The recorded payment-to-bill ratio is 90.12%. No payment dates or recognition rules are supplied, so these figures do not establish dated cash collection performance or recognised accounting revenue.

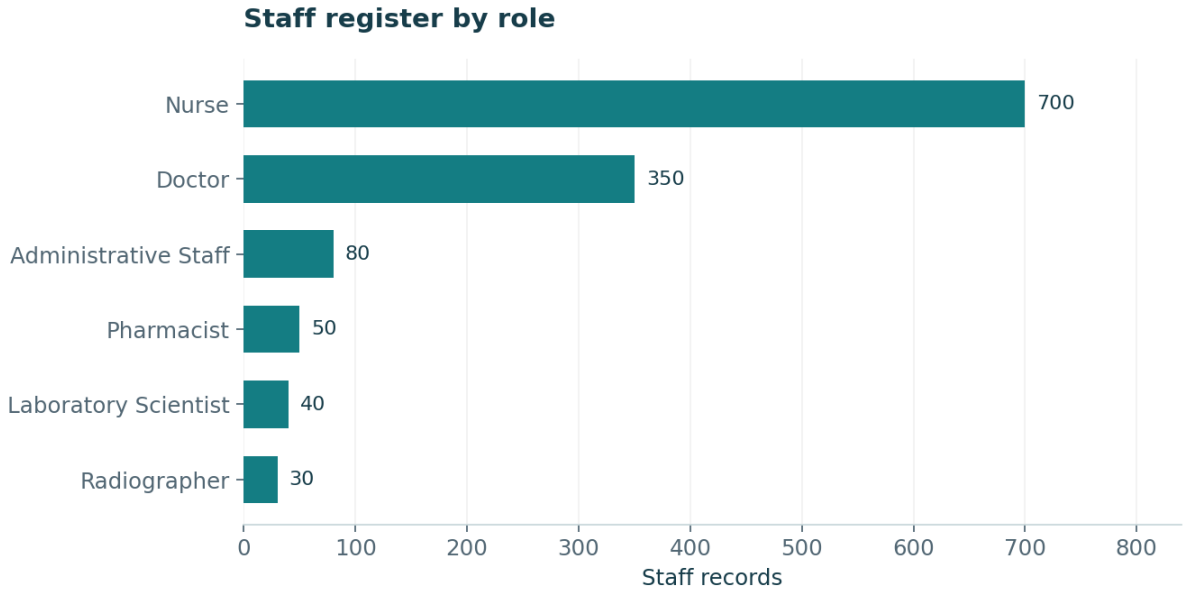
Source	Total	Limitation
Billing	RWF 9.309 billion	Invoice dates and ServiceID absent
Payment records	RWF 8.389 billion	Receipt dates absent; amounts kept independent of billing status
Revenue Summary	RWF 1.133 billion	Separate annual monthly series; not reconciled to transaction extracts
Insurance claims	42,036 records	21,028 Pending; 49.98% Approved share of all claims

Attribution and trend boundaries

Only the 5,000 appointment-linked bills support facility, department and patient-residence province comparisons. Revenue by service cannot be derived: neither billing nor appointments identifies ServiceID. Insurance claim providers identify claims and associated bills, not the patient master insurance-code dictionary. Monthly Revenue Summary values remain a separate unreconciled series; linked appointment-cohort amounts are not receipt-month cash flows.

Operations and patient experience

The staff extracts contain 1,250 records, including 350 doctors and 700 nurses. Clinical staff amount to 58.5 records per 1,000 registered patients. This ratio does not establish staffing adequacy: working hours, staffed capacity, case mix and complete encounter demand are not supplied.



Facility and capacity interpretation

Twenty facility identifiers appear across the files, but a facility master with names, types, ownership and geography is absent. Appointment FacilityID agrees with the referenced department's FacilityID in only 5.12% of rows. Activity uses the appointment's recorded facility; department IDs supply type labels without forcing a facility reassignment.

The inventory lists 2,000 beds, 400 rooms and 200 equipment items. All beds are labelled Available. Dated occupancy, bed assignments, room usage and equipment availability are absent; physical utilization and expansion requirements cannot be estimated.

Feedback findings

The all-response mean is 2.99/5. Ratings 1-2 represent 40.44% of responses; rating 5 represents 19.70%. Only 8.34% of feedback links to an available appointment. Facility and department rankings show their linked response counts and are descriptive comparisons of that subset. Monthly low-rating views follow appointment cohorts, not feedback-submission dates.

Executive recommendations

Rank	Action	Evidence and monitoring basis
1	Recover complete appointment extracts	55,000 of 60,000 bills lack an appointment match; monitor linkage coverage.
2	Reconcile financial source definitions	Resolve RWF 9.309bn billed, 8.389bn recorded payments and 1.133bn summary revenue.
3	Correct facility and department reference mappings	94.88% of appointment-to-department facility references disagree.
4	Publish a verified facility directory	Names, geography, ownership and facility types cannot currently be reported.
5	Add invoice dates, receipt dates and service keys	Enable dated balances, receipt-month trends and service-level attribution.
6	Review pending insurance claims	21,028 claims are Pending; track ageing and eventual resolution separately.
7	Investigate low-rating service feedback	40.44% of recorded ratings are 1-2; retain response counts and coverage.
8	Collect dated bed assignments and occupancy	A uniform Available label does not establish occupancy or bed utilization.
9	Maintain room and equipment usage histories	Capacity and availability indicators require actual operating observations.
10	Use controlled access for identifiable extracts	Keep patient contacts and individual clinical records out of public portfolio files.

Methods and reproducibility

Thirty canonical tables were prepared from 27 unchanged source files. Text was trimmed, numeric fields typed, and slash-form birth dates normalised to ISO dates. All supplied birth dates fall on 1 January; recorded Age is retained. Duplicate source copies were reconciled rather than appended. All 40 required SQL questions were executed in SQLite, which the programme team explicitly permitted. The pipeline passes 122 checks and 12 independent SQL reconciliations; the SQL backup was restored and its principal totals checked.

Sources: AnalystLab Africa Week 6 official brief and linked HealthConnect datasets; programme clarification permitting SQLite, 20 October 2026. Province boundaries: geoBoundaries / Rwanda Geo Portal, 2020, CC BY 4.0. Source links and field definitions accompany the project. No unavailable capacity, facility geography or clinical-outcome fields were invented.